Kelvin Jonathan Yoga

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Skills

Cryptography: Post-Quantum Cryptography (lattice- and hash-based), Zero-Knowledge Proofs (SNARKs, STARKs), Provable Security, Anonymous Credentials, Zero-Trust Architecture, PKI, Crypto-Agility, Coding Theory.
Implementation: Rust, C++ (C++23), C, Python; arkworks (Groth16), Plonky3, LaBRADOR, libiop; Git, Docker, CI/CD, Linux.

Education

University of New South Wales – BSc, Computer Science (Security Engineering) 2026
Grade: Distinction *AI X Creativity Hackathon (USyd): 1st of 15 teams with AI StoryVerse*

Experience

Applied Cryptographer – Clave Systems Jul 2026 – Nov 2026
- Collaborated in designing a secure authentication and data verification system with zero-trust assumptions.
- Implemented ZK-SNARK based cryptographic systems in Rust.
- Brought zero-knowledge and cryptography expertise to a policy-based system built on existing cryptographic schemes, contributing to a 70% improvement in delivery and production speed.
- Communicated SNARK/STARK requirements to align implementation with end goals.

PQ Cryptography Researcher – UNSW Apr 2026 – Present
- Research credential revocation for privacy-preserving (anonymous) credential systems under Dr. Sushmita Ruj.
- Design a revocation-friendly authenticated data structure for use inside zero-knowledge proofs, with formal security definitions and proofs.
- Build a Rust prototype; benchmark zero-knowledge proofs in Groth16, Plonky3, SmallWood and LaBRADOR.
- Writing a first-author paper (in preparation).

Security & Backend Engineer – Solar Central Mar 2026 – May 2026
- Developed Python REST APIs, SQL data access objects and input validation in an Agile team of 6.
- Ran code security audits and briefed the team on vulnerability risks and fixes.
- Built automated testing and CI/CD pipelines for reproducible deployments.

IT Support & Data Entry – Sydney Endocrinology and Diabetes Sep 2023 – Dec 2024
- Managed office hardware, the local network and data backups under strict privacy and access-control rules.

Open Source & Projects

RustCrypto – XMSS BDS Tree Traversal (Rust) PR #1470, under review
- Implemented BDS tree traversal for XMSS and multi-tree XMSS signing in RustCrypto's xmss crate, cutting signing from 1.9 s to 8.5 ms (height 10) and 123 s to 10.5 ms (height 16).
- Made stateful signing all-or-nothing (state advances on a copy, seed copies wiped on error), so a failed signature never moves the one-time key index.

Post-Quantum Ethereum: Hash-Based Multi-Signatures (C++23, team of 3)
- Implemented the Generalized XMSS scheme (IACR ePrint 2025/055) in C++23, ported from Rust.
- Built Winternitz and target-sum one-time signatures and tweakable hashes (SHA-256, BLAKE3).
- Designed the in-circuit verification for SNARK aggregation (encoding, Winternitz chains, Merkle path), used in an Aurora (libiop) prototype.

Reed-Solomon Error-Correcting Code (C++)
- Built an encoder and decoder over the prime field GF($2^{19} - 1$) with a Gao decoder, tested with GoogleTest.